



Predicting the risk of post-hepatectomy portal hypertension using a digital twin: A clinical proof of concept

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Background & Aims: Despite improvements in medical and surgical techniques, post-hepatectomy liver failure (PHLF) remains the leading cause of postoperative death. High postoperative portal vein pressure (P_{PV}) and portocaval gradient (PCG), which cannot be predicted by current tools, are the most important determinants of PHLF. Therefore, we aimed to evaluate a digital twin to predict the risk of postoperative portal hypertension (PHT).

Methods: We prospectively included 47 patients undergoing major hepatectomy. A mathematical (OD) model of the entire blood circulation was assessed and automatically calibrated from patient characteristics. Hepatic flows were obtained from pre-operative flow MRI ($n = 9$), intraoperative flowmetry ($n = 16$), or estimated from cardiac output ($n = 47$). Resection was then simulated in these 3 groups and the computed P_{PV} and PCG were compared to intraoperative data.

Results: Simulated post-hepatectomy pressures did not differ between the 3 groups, comparing well with collected data (no significant differences). In the entire cohort, the correlation between measured and simulated P_{PV} values was good ($r = 0.66$, no adjustment to intraoperative events) or excellent ($r = 0.75$) after adjustment, as well as for PCG (respectively $r = 0.59$ and $r = 0.80$). The difference between simulated and measured post-hepatectomy PCG was ≤ 3 mmHg in 96% of cases. Four patients suffered from lethal PHLF for whom the model satisfactorily predicted their postoperative pressures.

Conclusions: We demonstrated that a OD model could correctly anticipate postoperative PHT, even using estimated hepatic flow rates as input data. If this major conceptual step is confirmed, this algorithm could change our practice toward more tailor-made procedures, while ensuring satisfactory outcomes.

Lay summary: Post-hepatectomy portal hypertension is a major cause of liver failure and death, but no tool is available to accurately anticipate this potentially lethal complication for a given patient. Herein, we propose using a mathematical model to predict the portocaval gradient at the end of liver resection. We tested this model on a cohort of 47 patients undergoing major hepatectomy and demonstrated that it could modify current surgical decision-making algorithms.

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Introduction

The surgical and perioperative management of patients undergoing liver resection has changed dramatically over the last 30 years. Despite these improvements, postoperative mortality remains a considerable concern with mortality exceeding 10%.¹ Post-hepatectomy liver failure (PHLF) is the main cause of mortality,² largely explained by an excessive rise in portal vein pressure (P_{PV}) and the portocaval gradient (PCG) caused by a reduction of the parenchymal vascular bed which leads to higher intrahepatic resistance.³

The BCLC guidelines recently changed from a strict contraindication to resect in the event of pre-existing portal hypertension (PHT)⁴ towards recognizing a progressive risk dependent on the resection extent and the model for end-stage liver disease (MELD) score.⁵ However, although preoperative PHT is undoubtedly a risk factor for serious complications,⁶ its predictive value is poor^{7,8} and the most accurate predictor has been shown to be the post-resectional PCG value that can predict both PHLF and early mortality.^{9,10} Hence, the focus here is on post- rather than pre-resection liver pressures. Unfortunately, this variable cannot be anticipated using current tools.

With the development of numerical medicine, recent multi-parametric algorithms have attempted to predict postoperative outcomes^{11,12} but these calculators are not designed to reflect the patient-specific hemodynamic data. On the other hand, some models of the entire human circulation have been developed and some are emerging in the field of liver hemodynamics. While none has yet been used in routine practice, such modeling, which is mainly based on closed-loop OD simulations (a lumped

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parameter model, i.e. a compartmental model of the entire circulation, without any spatial dimension in equations) may efficiently reflect invasive measurement and pathophysiological situations.^{13,14} Other models have explored the consequences of partial hepatectomy.^{15,16} Our team recently reported a 0D closed-loop model tested on partial hepatectomies in pigs,¹⁷ followed by a clinical feasibility study.¹⁸ These promising results led us to think that this digital twin might assist our decision-making strategy and patient selection, and potentially avoid a surgery in the event of predictable high risk of postoperative PHT. Another advantage of such physiological models over statistical ones, is that they provide mechanistic interpretation of the causes of postoperative PHT for a specific patient or for inter-patient variability. They also enable quantification of hemodynamic variations caused by intraoperative events, which can play a role on top of the main surgical act.^{15,16}

Our aim was thus to apply our 0D model on a large-scale human series and to assess its accuracy in the setting of major hepatectomy, focused specifically on anticipating postoperative PCC.

Patients and methods

Clinical and paraclinical features

Study population

Forty-seven non-consecutive patients resected in our tertiary center were prospectively included and followed up until post-operative day 90. The patients were informed and gave their consent to participate to this study, which had been approved by the local Ethics Committee (ID RCB: 2018-A03316-49). The inclusion criteria were open major hepatectomy with access to intraoperative measurements of vascular pressures and anesthesiologic parameters. No clinical or surgical decisions were based on the results of the numerical simulations, which were performed retrospectively.

Preoperative imaging and preparation

All patients underwent a contrast CT scan during the month prior to surgery. Liver volumetry was performed by a single operator (NG) using Myrian XP-Liver (Intrasense, Montpellier, France). In the event of an indication for portal vein embolization (PVE), the liver volumes used in the algorithm were considered after PVE. Some patients ($n = 9$) underwent a flow MRI within the preoperative month to quantify cardiac output (CO), as well as portal vein (PV) and hepatic artery (HA) flows [Q_{PV} , Q_{HA}]. Details about MRI examinations are provided in the [supplementary data](#).

Intraoperative measurements to run or validate the algorithm

For all 47 patients (excepted specific indication), all measurements were collected just before and then after hepatectomy (i.e. after hemodynamic stabilization, immediately before parietal closure): Heart rate (HR); CO estimated from the thermodilution technique or pulse contour analysis; mean arterial pressure (P_{AO}); Inferior vena cava (IVC) and PV pressures measured by a transducer connected to a needle inserted into the vessels (pressure measurements were taken twice in all patients); Q_{PV} and Q_{HA} measured in 16 patients using a transit-time ultrasound flowmeter Transonic HT313; volume of blood loss and total volume infused; administration of a vasoconstrictor. Post-operative outcomes, including PHLF,^{19,20} were collected up to postoperative day 30.

Mathematical model and parametrization

Mathematical model of the entire blood circulation

The model has been described elsewhere in detail.^{17,18,21} The human cardiovascular system can be seen as a hydraulic network (the vessels), driven by a power source (the heart). This analogy identifies fluid flow with an electrical intensity and pressure with voltage, which enables modeling of the entire circulation like an electrical circuit. Viscous dissipation through vessels and valves is modeled by resistances, while electronic capacitances (or elastances) represent vessel and cardiac compliance.

The model developed for partial hepatectomy is able to represent dual liver perfusion (arterial + venous systems) and its interaction with the rest of the cardiovascular system. Our previous scheme was adapted to human anatomy ([Fig. 1](#)): the liver was modeled as 2 parallel components (2 hemi-livers), whose parameters (namely resistances in the portal network, arterial network and liver tissue: R_{PV} , R_{HA} , and R_l) were indexed on the mass of each hemi-liver. In addition to the cardiopulmonary system, it included 3 blocks: the digestive organs, the liver, and a single block representing all other organs. The pulsatile activity of the heart was represented by periodic time-varying elastances $E(t)$ with 2 parameters for each of the 4 cardiac contractile chambers, as well as a nonlinear valve function for each cardiac valve. Overall, the model produced a system of 32 differential-algebraic equations.

Procedure for model calibration (see [supplementary information](#) for more details)

In the pre-hepatectomy stage, the model needs to reflect the patient's state. The model parameters were set such that the model variables are close to the measured data ([Fig. 1](#) right) for each patient. To compensate for the limited patient data available to set all model parameters, assumptions were also taken from the literature.^{17,21} In a first step, all the systemic and pulmonary parameters were estimated. Then, the heart block

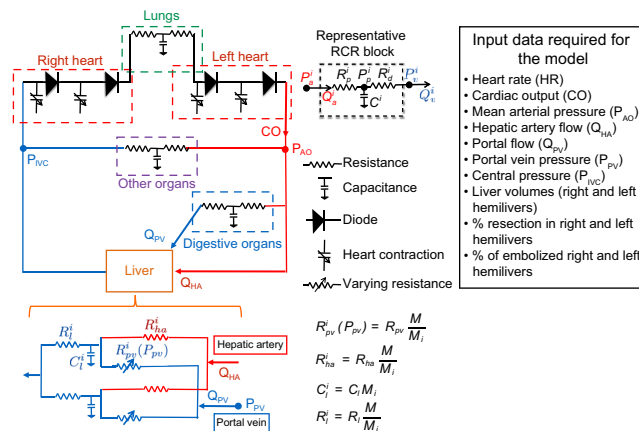


Fig. 1. Schematic representation of the 0D closed-loop cardiovascular model for humans (2 hemi-livers). A complete simulation includes 2 successive steps: i) A patient-specific calibration according to pre-resection measurements (black box showing the required variables) and data from the literature: each electrical component is adjusted to the patient's characteristics; ii) A simulation of hepatectomy according to the percentage of each hemi-liver resected, eventually adjusting to blood loss/infusion, variations in heart rate and the administration of vasoconstrictors as well as vessel stiffening in the case of fibrotic livers. (This figure appears in color on the web)

parameters were set based on HR and an optimization algorithm (covariance matrix adaptation evolution strategy²²), so that the model variables matched the best 6 of a patient's major hemodynamic measures: CO, P_{AO} , P_{PV} , PCG ($=P_{PV}-P_{IVC}$), Q_{PV} and Q_{HA} .

In the event of PVE ($n = 12$), the system was tuned according to the pre-resection data, taking account of any pre-existing absence of right (total or sectorial) portal flow.

This calibration step requires flow and pressure measurements before resection. In this work we propose 2 possibilities for flow measurements (*Flit* and *Fmri*) that will be compared and verified with intraoperative measurements before resection (*Fflowmet*). These 3 different types of input data were considered and compared (Fig. 2) in view of further preoperative and non-invasive use of the algorithm:

- *Fflowmet*: Pre-resectional values measured as Q_{PV} and Q_{HA} through flowmeter ($n = 16$),
- *Flit*: Estimated flows = Q_{PV} and Q_{HA} were considered from the literature as being 20% and 5% of the pre-resectional CO values provided by anesthesiologists^{23,24} ($n = 47$),
- *Fmri*: Q_{PV} , Q_{HA} and CO provided by preoperative flow MRI ($n = 9$, of which 5 also underwent complete intraoperative flowmetry measures).

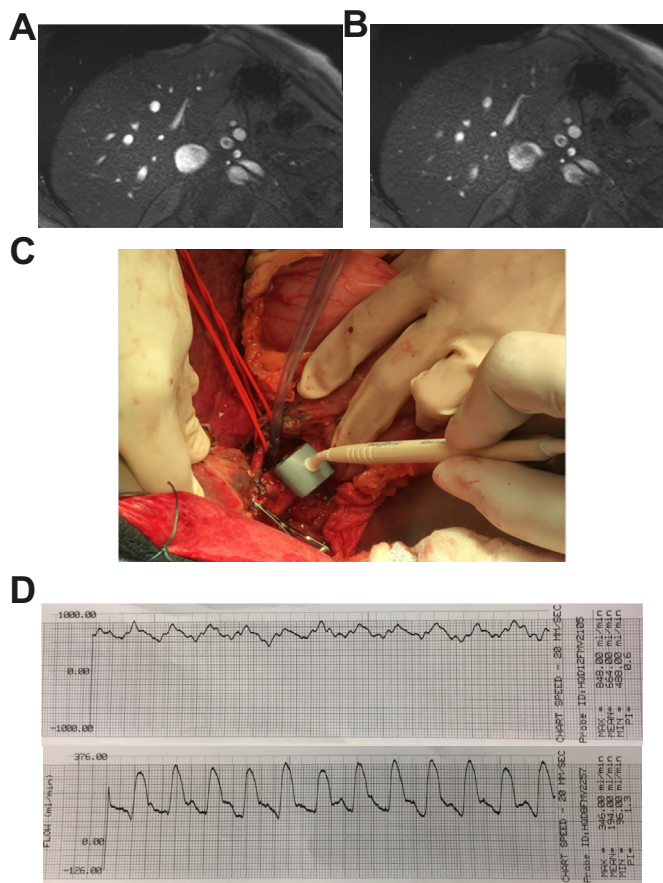


Fig. 2. Patient-specific flow measures used to calibrate the model provided by flow MRI or intraoperative flowmetry. Input data to run the 0D algorithm were obtained from flow MRI (A and B, same patient at different cardiac times) or intraoperative flowmetry (C) providing portal (D, up) and arterial (D, down) liver flow estimates. (This figure appears in color on the web)

Virtual hepatectomy

After optimizing the parameters according to preoperative patient data, hepatectomy was simulated by decreasing the mass of the resected part to 0 (Fig. 3). A complete simulation lasted about 45 minutes.

Furthermore, during surgery, the hemodynamic status of the patient may change. Thus, an “adjusted” simulation was also performed that considered intraoperative data, i.e. the HR after resection, variations in blood volume and potential drug-induced vasoconstriction or surgery-induced vasodilation. Note that only “unadjusted” virtual hepatectomy simulations could be run, with the current methodology, before resection.

For adjusted simulations, in the case of a fibrotic liver (fibrosis $\geq F2$ Metavir), we considered that the portal vessels cannot adjust to an increase of pressure (rigid vessels) whereas in all other cases (normal or weakly fibrotic livers, as well as all unadjusted simulations), the PV resistance depended on P_{PV} .¹⁷

Statistical analysis

The primary objective of the study was to predict the post-operative PCG in the whole cohort with a precision of ± 3 mmHg (choice compatible with clinical decision-making and intraoperative precision of instruments), using estimated flow data (*Flit*) as the input. The secondary objectives were to analyze the accuracy of pressure simulations, and to compare the simulation results according to the input data (*Fmri*, *Flit* or *Fflowmet*). The sample size calculation is justified in the [supplementary information](#).

Results

Population characteristics, perioperative features

We included 47 patients (gender ratio M/F: 55%/45%), with a mean age of 63 ± 12 years. The indication for resection was malignant disease in 41 cases (87%): cholangiocarcinoma ($n = 24$), hepatocellular carcinoma ($n = 11$) and colorectal metastases ($n = 6$). The mean MELD score was of 7.9 ± 2.1 , and the mean platelet count was $260 \pm 86 \times 10^3/L$. A fibrotic ($\geq F1$ fibrosis) and cirrhotic (F4)

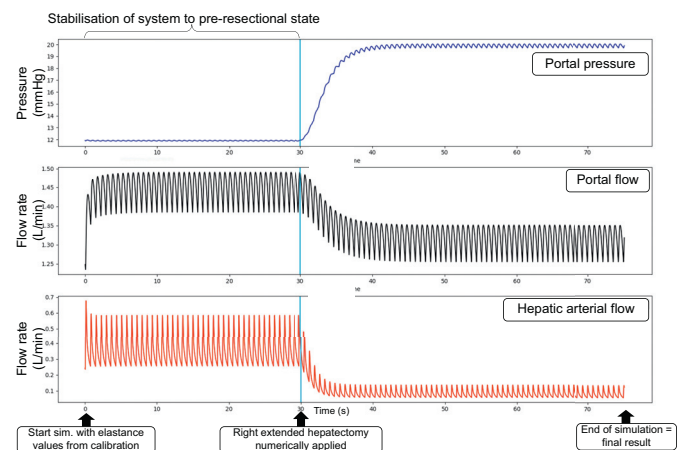


Fig. 3. Example of results provided by a 0D simulation. Example of a complete simulation of left hepatectomy in terms of changes to portal pressure, portal flow and hepatic artery flow. Hepatectomy is simulated numerically at 30 seconds, leading to an increase in portal pressure and a decrease in hepatic artery flow. Portal flow decreases slightly. (This figure appears in color on the web)

parenchyma were observed in 10 (21%) and 3 (6%) cases, respectively. The median MELD score of cirrhotic patients was 9. The mean number of resected segments was 4.7 ± 1.2 (median percentage of resected volume: 58.7%, range: 16–89%). Five patients (10.6%) died postoperatively, following PHLF ($n = 4$) or extrahepatic complications ($n = 1$). Concerning the 4 PHLF-related deaths (3 efficiently drained perihilar cholangiocarcinoma + 1 hepatocellular carcinoma/cirrhosis), all underwent preoperative PVE (2 complete, 2 uncomplete [2 segments only]) and all of them presented a future remnant liver volume to body weight ratio higher than 0.5% (median: 0.9%; range: 0.64–1.1) with a median percentage of resected volume of 58.8% (range: 40–69%, right hepatectomy ($n = 1$) or right extended to segments 1+4 [$n = 3$]). Their median pre-resectional P_{PV} and PCG values were 16 (11–22) mmHg and 9 (8–11) mmHg respectively, while their post-hepatectomy P_{PV} was 23.5 (18–30) mmHg and PCG was 17.5 (14–20) mmHg. Among the 4 patients with PHLF, 2 presented a pre-resectional PCG ≥ 10 mmHg.

Simulations on the whole cohort based on estimated flows (Flit, no invasive flowmeter measurements).

Thirty-six patients (77%) experienced intraoperative events which meant that adjusted simulations differed from unadjusted simulations. In 14 (39%) of these cases, this modification only concerned a variation in blood volume.

As shown in Fig. 4A, the predictions of post-resectional P_{PV} and PCG were quite accurate and indeed improved slightly when intraoperative events were considered. There was no significant difference between measured and simulated values for either variable.

The correlation between measured and simulated P_{PV} values was moderate ($r = 0.66$, $p < 0.0001$, no adjustment to intraoperative events) or strong ($r = 0.75$, $p < 0.0001$) after adjusting for hemodynamic modifications (Fig. 4B) and for PCG ($r = 0.59$ and $r = 0.80$, respectively, $p < 0.0001$). Concerning the simulated PCG, the performance of the OD model to predict a post-hepatectomy PCG value ≥ 10 mmHg was:

- Unadjusted simulations: sensitivity = 69%, specificity = 94%, positive predictive value = 82%, negative predictive value = 89%,
- Adjusted simulations: sensitivity = 100%, specificity = 94%, positive predictive value = 87%, negative predictive value = 100%.

Unadjusted simulations enabled an accuracy of 72% within the primary endpoint and with a calculated bias of -0.72 mmHg. As illustrated in Fig. 4C, 2 patients presented an adjusted simulated PCG that was outside the primary objective (absolute values of the difference between the simulation and measurement: 3.2 and 5.1 mmHg), meaning that 95.7% of adjusted predictions were within the ± 3 mmHg margin. The calculated bias was -0.05 mmHg. These 2 patients underwent almost bloodless left hepatectomy on non-cirrhotic liver. In the first patient, the measured PCG fell by 3 mmHg between pre- and post-resection, whereas the simulation showed a slight increase in the gradient as might be expected in such a situation with no significant intraoperative event, hinting at a measurement error. In the second case, the measured PCG value rose from 1 to 7 mmHg after resection, without any clear explanation, whereas the simulation only predicted a PCG increase of 1 mmHg; this example highlights the uncertainty of pre-resection measurements when the PCG is low.

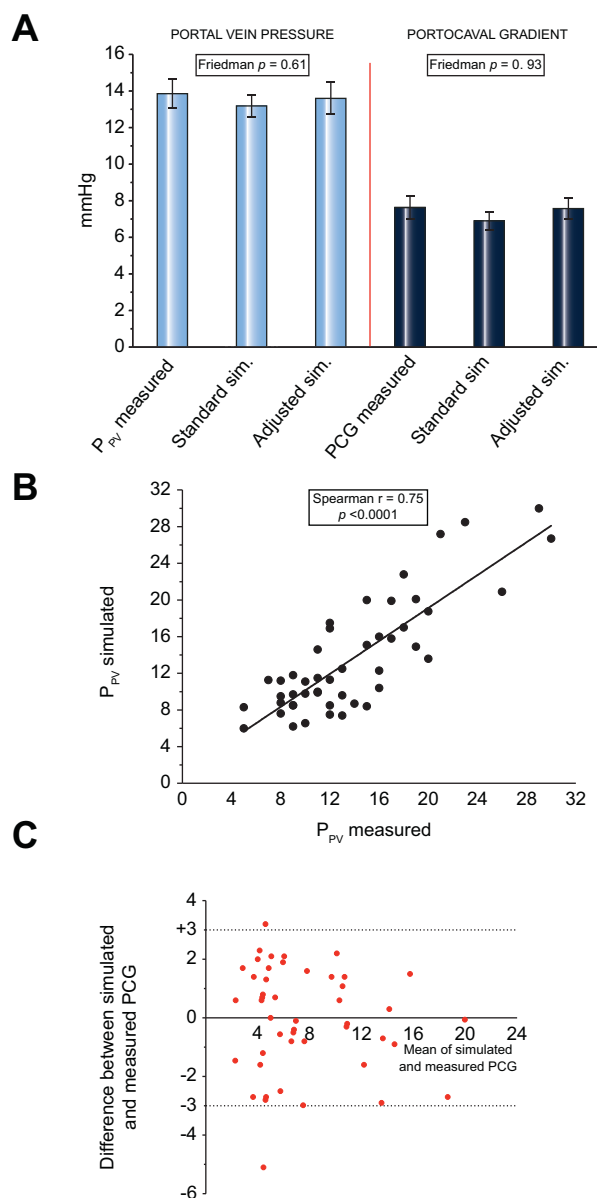


Fig. 4. Comparison between simulated and measured post-hepatectomy pressures on the whole cohort. The Flit (Q_{PV} and Q_{HA} considered as being 20% and 5% of cardiac output, respectively) simulation results are presented. Adjusted simulations considered intraoperative hemodynamic events (i.e. modifications to HR, blood loss and vascular vasoconstriction/vasodilation). (A) Comparison between measured and simulated P_{PV} ($p = 0.61$, Friedman test) and PCG ($p = 0.93$, Friedman test). While statistically non-significant, adjusted simulations tended to be more accurate than standard ones. Graphs represent mean $\pm$ standard error of the mean. (B) correlation between measured and simulated P_{PV} ($p < 0.0001$, $r = 0.75$, Spearman test). Pressure is expressed in mmHg. (C) Bland-Altman analysis of measured and adjusted simulated PCG (in mmHg). 95.7% of simulations are within the range: measurement ± 3 mmHg. Flit, calibration of the algorithm is based on liver flow estimates from the intraoperative cardiac output (repartition % from the literature); HA, hepatic artery; HR, heart rate; PCG, portocaval gradient; P_{PV} , portal vein pressure; PV, portal vein; Q, flow. (This figure appears in color on the web)

The 4 patients who died postoperatively from PHLF had simulated post-hepatectomy PCG values within this margin of error, with a median adjusted predicted PCG of 16.9 mmHg

(range = 13.3–20 mmHg). Using a ROC (receiver-operating characteristic) analysis, we determined that the best cut-off points for post-resectional measured P_{PV} and PCG values to anticipate PHLF were 17.5 mmHg (area under the curve = 0.94, $p = 0.004$, sensitivity = 100%, specificity = 84%) and 13.5 mmHg (area under curve = 0.98, $p = 0.002$, sensitivity = 100%, specificity = 93%), respectively. Among these 4 deceased patients, 3 presented an unadjusted predicted P_{PV} that was higher than 17.5 mmHg.

Comparison of pre-resectional flow data according to 3 measuring methods

By comparing the Q_{PV} and Q_{HA} flow values provided by MRI (Fmri), intraoperative flowmeter (Fflowmet) and estimated measurements (Flit), there was no statistically significant difference ($p > 0.15$ for each comparison; Wilcoxon matched pairs tests). In the Fflowmet group, median Q_{PV} and Q_{HA} represented 15.3% (3–26) and 3.6% (1–7) of total CO, and respectively 82% (40–93) and 18% (7–59) of total hepatic flow.

Calibrations and simulations from flows measured through a flowmeter vs. values estimated from the literature as input data: Does it matter?

Secondary results comparing post-resectional simulations obtained using real flow measured as input for the algorithm, vs. literature-based estimated flows. This aimed to understand if patient-specific liver flow data led to more accurate simulations than literature-based flow data.

As shown in Fig. 5C, the preoperative calibration did not differ between the 2 types of input flow data ($p = 0.93$). By applying the Fflowmet or Flit as input for the algorithm calibration, we reached highly significant correlations for P_{PV} and PCG ($r = 0.99$, $p < 0.0001$) between the simulated variables and patient measurements, whatever the input chosen (Fig. S1). This means that in both cases, calibration of the model was successful in reflecting the patient's state prior to resection. When performing virtual hepatectomy, model predictions for the unadjusted simulations (Fig. 6C) did not differ between Fflowmet and Flit with respect to post-resectional P_{PV} ($p = 0.82$) or PCG ($p = 0.54$).

Simulations with the MRI flows as input data

Secondary results on post-resectional simulations based on flow MRI data as input for the algorithm, with the view of later pre-operative clinical use for decisional algorithm.

The CO and HR data from preoperative MRI and intra-operative pre-resectional anesthesiologic measurements were also comparable ($p = 1$ and $p = 0.69$, respectively).

The calibrations based on Fmri or Flit ($n = 9$) were not statistically different ($p = 0.94$), and the same applied to Fmri vs. Fflowmet ($n = 5$, $p = 0.31$) (Fig. 5). The unadjusted simulations of post-hepatectomy status based on Fmri or Flit, as well as those based on Fmri vs. Fflowmet were not statistically different with respect to P_{PV} and PCG (Fig. 6A, 6B).

Discussion

Statement of principal findings

This feasibility study is the first application in humans of a 0D model of the entire blood circulation, applied in the setting of liver resection. In 47 patients, we succeeded to anticipate post-resectional P_{PV} and PCG values, 2 well-known predictors of PHLF,^{9,25} with a reasonable correlation, even irrespective of intraoperative events.

We showed that it was possible to run the algorithm with various input liver flows, and to reach equivalent accuracy for calibration and simulation.

This study also led us to propose post-resectional cut-offs for P_{PV} (17.5 mmHg) and PCG (13.5 mmHg) that predict PHLF. This is a novel finding, as no other team has reported such information to date. In 2013, we reported a slightly higher P_{PV} cut-off (20 mmHg); this different finding may be explained by our current population cohort including cirrhotic patients (unlike Allard M.A. et al.⁹) more prone to develop liver failure at a lower P_{PV} level.

The 4 patients who died of PHLF presented with a preoperative future remnant liver volume ratio of over 0.5% body weight, usually considered to be sufficient for resection on non-cirrhotic livers.²⁶ This confirms that our digital twin might provide other selection criteria that enable safe resection, since the simulated (adjusted) postoperative PCG values of these patients were alarming and would have contraindicated the procedure.

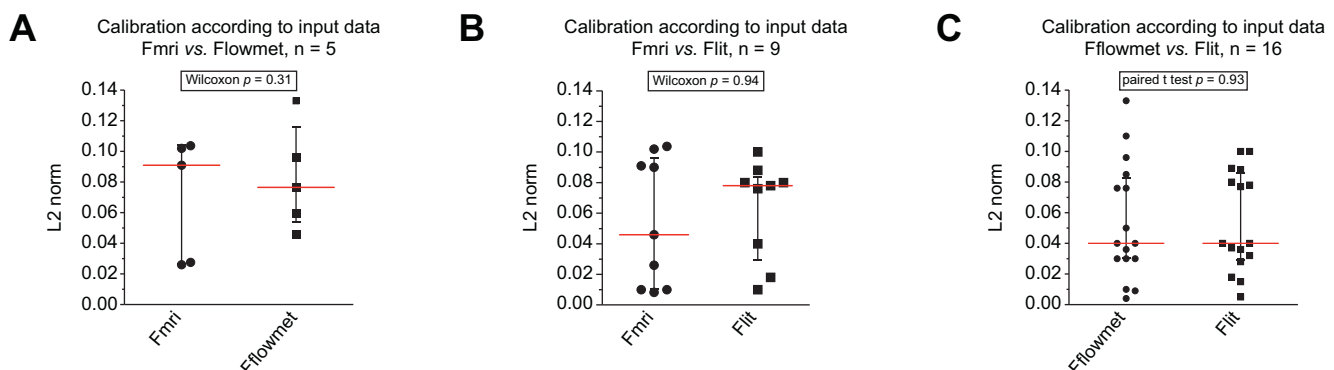


Fig. 5. Calibration results (1st step of simulation). The calibration, expressed as the L2 norm and representing a composite score for calibration efficiency, was statistically equivalent in all input situations (a low L2 norm represents an accurate calibration with simulated variables almost equal to measurements). Graphs represent medians with interquartile ranges. Results of calibration are drawn as a function of input data: (A) Fmri vs. Fflowmet ($n = 5$). Wilcoxon test, $p = 0.31$; (B) Fmri vs. Flit ($n = 9$, Q_{PV} and Q_{HA} considered as being 20% and 5% of measured cardiac output, respectively). Wilcoxon test, $p = 0.94$; (C) Fflowmet vs. Flit ($n = 16$). Paired t test, $p = 0.93$. Fflowmet, calibration of the algorithm is based on intraoperative flow measurements obtained using a flowmeter; Flit, calibration of the algorithm is based on liver flow estimates from the intraoperative cardiac output (repartition % from the literature); Fmri, calibration of the algorithm is based on preoperative flow measurements from MRI; HA, hepatic artery; PV, portal vein; Q, flow. (This figure appears in color on the web.)

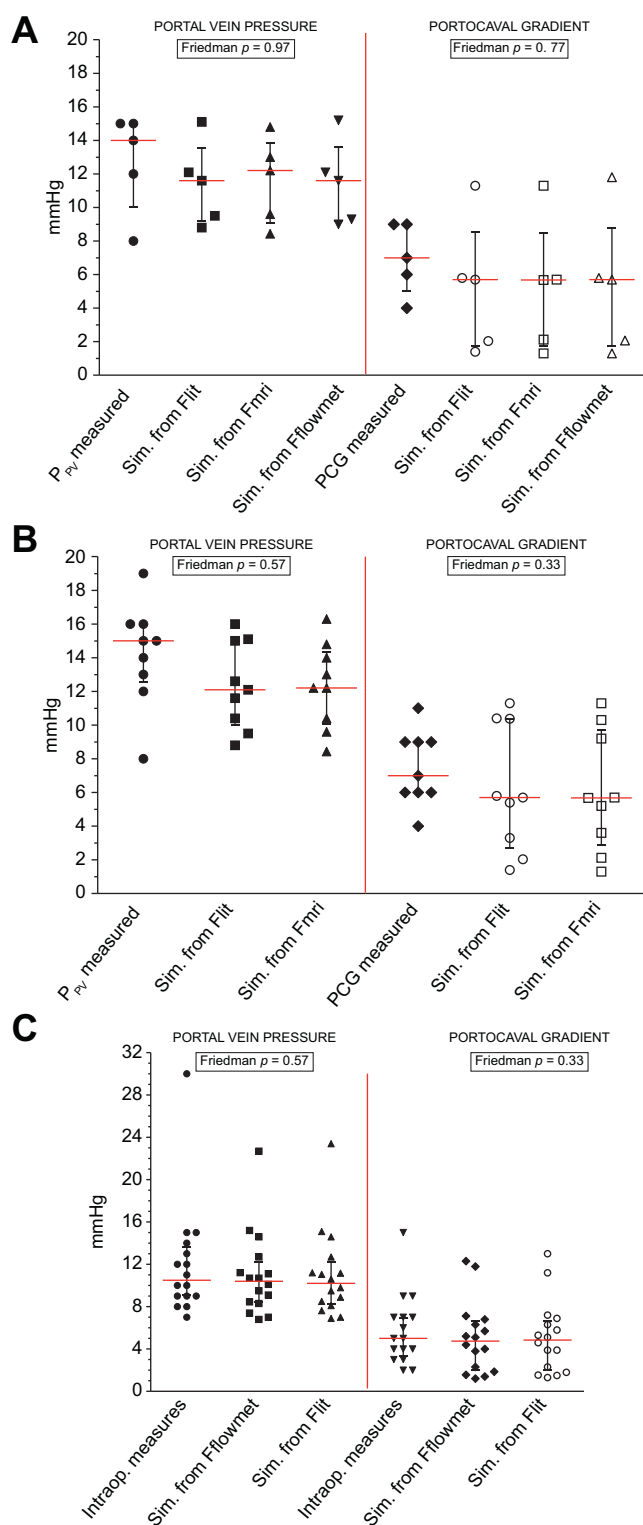


Fig. 6. Comparison between simulated and measured post-hepatectomy pressures in subgroups. Results of simulations as a function of input data: *Fmri* vs. *Fflowmet* vs. *Flit* (Q_{PV} and Q_{HA} considered as being 20% and 5% of measured cardiac output, respectively). Graphs represent medians with inter-quartile range. Friedman tests used for all the analyses. (A) *Flit* vs. *Fflowmet* ($n = 5$): no statistically significant difference for post-resection P_{PV} ($p = 0.97$) and simulated PCG ($p = 0.77$); (B) *Flit* vs. *Fmri* ($n = 9$): no statistically significant difference for post-resection P_{PV} ($p = 0.57$) and simulated PCG ($p = 0.33$); (C) *Fflowmet* vs. *Flit* ($n = 16$): no statistically significant difference for post-

Justification of the endpoint and of the need for such a model

We aimed to predict post-resectional P_{PV} and PCG because, until now, no tool could allow such predictions while these variables consensually appear as a major risk factor of PHLF. Roayaie *et al.* clearly demonstrated that recent guidelines were unable to provide any fair algorithm to select ideal candidates for resection, particularly based on preoperative PHT level.²⁷ Conversely, using our numerical tool based on patient-specific parameters, we were able to precisely model the risk of post-resectional PHT. This finding highlights the need for such a physiologically complex model, which by design considers several interplaying factors.

Our model accurately predicted P_{PV} and PCG variations, which are obviously linked. However, because of a possible bias of P_{PV} as a function of central pressure,²⁸ we preferred to focus on PCG, a more reliable and reproducible measure. In fact, in the event of an inaccurate absolute calibration of the instruments used, the PCG remains reliable whereas P_{PV} is not.

One could argue that a clinical primary endpoint would have been more useful for clinicians. In fact, to predict a clinical condition related to PHT (ascites, death) or a biological event (liver failure) would be interesting but, from a statistical point of view, it was not possible to assess the accuracy of our model, by comparing it to a such a rare event. By contrast, all patients had postoperative measures available to compare our simulations results with.

Interpretation with reference to other studies

For several decades, preoperative PHT ($PCG \geq 10$ mmHg) was considered as the principal criterion to contraindicate liver resection.²⁹ We now know that, even if the degree of pre-resectional PHT is available, it is a poorer predictor of PHLF than the post-resectional P_{PV} value⁹ or even a variation in P_{PV} .²⁵ This justifies the need for a more physiologically realistic tool to prevent PHLF and avoid unnecessarily excluding patients from potentially curable surgery.⁷ As a proof of that, 50% of our patients with preoperative $PCG \geq 10$ mmHg had simple outcomes.

Because of these concerns regarding the predictive value of invasive or non-invasive³⁰ preoperative PCG alone to select patients for surgery, we developed a digital twin model applicable to medicine. This OD model appears to be predictive of hemodynamic fluctuations due to hepatectomy, since we were able to report here on increases of P_{PV} of 23% and 24% (unadjusted and adjusted simulations, respectively), which is consistent with the findings of Bogner *et al.* who reported that mean P_{PV} values rose by 27% after major hepatectomy.²⁵

Some OD hemodynamic models have already been reported in the field of hepatology, but no work has included predictive and patient-specific simulations. It was therefore not possible to compare our model's performance to others.

resection P_{PV} ($p = 0.82$) and simulated PCG ($p = 0.54$). *Fflowmet*, calibration of the algorithm is based on intraoperative flow measurements obtained using a flowmeter; *Flit*, calibration of the algorithm is based on liver flow estimates from the intraoperative cardiac output (repartition % from the literature); *Fmri*, calibration of the algorithm is based on preoperative flow measurements from MRI; PCG, portocaval gradient; P_{PV} , portal vein pressure. (This figure appears in color on the web.)

Strengths and weaknesses of the study

The main strength of this feasibility study was the large sample size tested and the good performance of the predictions from a model that answers to a daily clinical issue. Knowing that the time required for a single simulation is compatible with everyday practice (<1 hour), our findings may largely modify the selection criteria used in routine practice and improve patient outcomes.

Regarding the difference of accuracy between adjusted and unadjusted simulations, we should bear in mind that correlations between unadjusted simulations and measured data need to be interpreted cautiously since post-resectional measures are, by definition, the consequences of intraoperative events that are not considered in unadjusted simulations. In fact, the unadjusted simulations represent the effect of the surgical act alone. Unadjusted simulations may also reflect the delayed (mid-term) more stable hemodynamics conditions, whereas intraoperative measures reflect the immediate status after anesthesiologic maneuvers to compensate for changes in circulating blood volume.

Some limitations of the study must be highlighted. The small sample size of subgroups (*Fflowmet* and *Fmri*) must be kept in mind and encourage us to interpret cautiously statistical results. However, most of the results presented no statistical trend and a larger cohort may not have brought any additional significant result.

In the *Flit* group, liver flows were not modulated according to the size of hepatocellular carcinoma (the 5% CO rate chosen could be increased in the case of hypervascularized tumors) or the degree of hepatopathy ($Q_{PV} = 20\%$ of CO could be an overestimate in the event of fibrosis), so that even better results might be expected following a case-by-case adjustment of estimated flows.³¹ However, the ratio we chose ($Q_{PV} = 4 \times Q_{HA}$) was validated by the flow rate observed in the *Fflowmet* group (ratio: 4.5).

Finally, we did not integrate liver stiffness into our OD model (because of lacking data in non-cirrhotic patients) even though this parameter is correlated with PCG³² and postoperative outcomes.^{33,34} This may offer a new approach for more refined development, particularly because, to date, we have adjusted PV elasticity to the fibrosis grade shown in the pathological report, but this information is not always available prior to surgery. Integrating liver stiffness could therefore improve the accuracy of simulation in terms of patient selection.

Further development and use of this OD model

The current work may be considered as a pilot study necessary to confirm the relevance and feasibility of this model in humans. So far, all the simulations were performed postoperatively, and we now need to assess the clinical impact of such a tool in a prospective fashion. In fact, any retrospective or external validation could not be easily applicable as many input data could be lacking. To this end, acquisition data are ongoing and we will analyze on a larger scale different inputs to the algorithm: i) Flow MRI phases may be directly acquired during standard MRI for every potential surgical candidate to get the flows; ii) The combination of CO data alone (easily available from a flow MRI or cardiac sonography performed for anesthetic consultation) with PV/HA flows extrapolated from the literature (20%/5% or another breakdown adapted to tumor size and/or liver stiffness) might facilitate routine use of our model; iii) Pressure data obtained from transjugular approach (wedged

and free hepatic venous pressure³⁵) in order to run the model even in laparoscopy without any access to intraoperative pressure measures; iv) A population including cirrhotic patients requiring limited (minor) resection should be tested as well, as we know that even small volume resected may lead to severe PHT and liver failure.³⁶

With these inputs, we will be able to use the model preoperatively as a clinical-decision tool (=tailored and digital-guided strategy) in order to quantify if the standard decisional-tree needs to be modified (i.e. extent of resection, need for portal flow modulation, recused surgery...), to what extent, and to assess the outcomes of so-called "extended" indications for liver resection.

Conclusion

We have demonstrated that a OD model of the entire blood circulation could be calibrated to reflect preoperative conditions, and accurately predict postoperative hemodynamic status. The unadjusted simulations may reflect the state of the patient when back to baseline hemodynamics conditions. When taking account of intraoperative events, the model was able to correctly predict postoperative PCG values, particularly in the 4 patients who died from PHLF, although their preoperative future remnant liver volume ratios were considered sufficient. In fact, the extent of liver resection alone did not correlate with portal pressure or PCG increase, highlighting the need for a physiological numerical model that naturally integrates the interplay of liver quality, extent of hepatectomy and the rest of the circulation. If confirmed by broader prospective studies, this algorithm might crucially individualize our practice. In the near future, if this digital twin is run preoperatively, surgeons could reject patients predicted to have an excessive risk of PHT, or confirm resection in otherwise borderline patients, and ensure satisfactory outcomes. The present OD model is a concrete example of a numerically assisted clinical tool that could move practice from empirically based medicine to a new digital era of enhanced surgery. It not only illustrates that individual outcomes can be predicted, but also confirms that complex hemodynamic systems can be simplified and numerically modeled to create useful digital twins. This model could be considered as "augmented intelligence" as it could assist surgeons with their judgments regarding patient selection and therapeutic indications, reducing postoperative morbi-mortality.

Abbreviations

CO, cardiac output; Ea, amplitude of elastance; Eb, basal elastance; *Fflowmet*, calibration of the algorithm is based on intraoperative flow measurements obtained using a flowmeter; *Flit*, calibration of the algorithm is based on liver flow estimates from the intraoperative cardiac output (repartition % from the literature); *Fmri*, calibration of the algorithm is based on preoperative flow measurements from MRI; HA, hepatic artery; HR, heart rate; IVC, inferior vena cava; P, pressure; PCG, portocaval gradient; PHLF, post-hepatectomy liver failure; PHT, portal hypertension; P_{PV} , portal vein pressure; PV, portal vein; PVE, portal vein embolization; Q, flow rate; R, resistance.

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Conflict of interest

The authors declare no conflicts of interest that pertain to this work.

Please refer to the accompanying [ICMJE disclosure forms](#) for further details.

Authors' contributions

Golse: investigation, analysis, methodology, writing manuscript. Joly: software creation, technical advice. Combari: investigation, analysis. Lewin: data collection. Nicolas: software creation. Audebert: software creation, methodology. Allard: data collection, analysis. Sa Cunha, Castaing, Cherqui, Adam: data collection, review manuscript. Samuel: corrected manuscript. Vibert and Vignon-Clementel: project managers, methodology, writing and review manuscript.

Data availability statement

All the data may be available on request to corresponding author.

Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhep.2020.10.036>.

References

- [1] Filmann N, Walter D, Schadde E, Bruns C, Keck T, Lang H, et al. Mortality after liver surgery in Germany. *Br J Surg* 2019;106(11):1523–1529.
- [2] Vibert E, Pittau G, Gelli M, Sa Cunha A, Jamot L, Faivre J, et al. Actual incidence and long-term consequences of posthepatectomy liver failure after hepatectomy for colorectal liver metastases. *Surgery* 2014;155(1):94–105.
- [3] Golse N, Bucur PO, Adam R, Castaing D, Sa Cunha A, Vibert E. New paradigms in post-hepatectomy liver failure. *J Gastrointest Surg* 2013;17(3):593–605.
- [4] European Association For The Study Of The Liver, European Organisation For Research And Treatment Of Cancer. EASL-EORTC clinical practice guidelines: management of hepatocellular carcinoma. *J Hepatol* 2012;56:908–943.
- [5] European Association for the Study of the Liver, European Association For The Study Of The Liver. EASL clinical practice guidelines: management of hepatocellular carcinoma. *J Hepatol* 2018;69(1):182–236.
- [6] Choi SB, Kim HJ, Song TJ, Ahn HS, Choi SY. Influence of clinically significant portal hypertension on surgical outcomes and survival following hepatectomy for hepatocellular carcinoma: a systematic review and meta-analysis. *J Hepatobiliary Pancreat Sci* 2014;21(9):639–647.
- [7] Cucchetti A, Cescon M, Golfieri R, Piscaglia F, Renzulli M, Neri F, et al. Hepatic venous pressure gradient in the preoperative assessment of patients with resectable hepatocellular carcinoma. *J Hepatol* 2016;64(1):79–86.
- [8] Boleslawski E, Petrovai G, Truant S, Dharancy S, Duhamel A, Salleron J, et al. Hepatic venous pressure gradient in the assessment of portal hypertension before liver resection in patients with cirrhosis. *Br J Surg* 2012;99(6):855–863.
- [9] Allard MA, Adam R, Bucur PO, Termos S, Sa Cunha A, Bismuth H, et al. Posthepatectomy portal vein pressure predicts liver failure and mortality after major liver resection on noncirrhotic liver. *Ann Surg* 2013;258(5):822–829. discussion 829–30.
- [10] Rhaïem R, Piardi T, Chetboun M, Pessaux P, Lestra T, Memeo R, et al. Portal inflow modulation by somatostatin after major Liver resection: a pilot study. *Ann Surg* 2018;267(6):e101–e103.
- [11] Berardi G, Morise Z, Sposito C, Igarashi K, Panetta V, Simonelli I, et al. Development of a nomogram to predict outcome after liver resection for hepatocellular carcinoma in Child-Pugh B cirrhosis. *J Hepatol* 2020;72(1):75–84.
- [12] Prodeau M, Drumez E, Duhamel A, Vibert E, Farges O, Lassailly G, et al. An ordinal model to predict the risk of symptomatic liver failure in patients with cirrhosis undergoing hepatectomy. *J Hepatol* 2020;71(5):920–929.
- [13] Wang T, Liang F, Zhou Z, Qi X. Global sensitivity analysis of hepatic venous pressure gradient (HVPG) measurement with a stochastic computational model of the hepatic circulation. *Comput Biol Med* 2018;97:124–136.
- [14] Audebert C, Peeters G, Segers P, Laleman W, Monbaliu D, Korf H, et al. Closed-loop lumped parameter modeling of hemodynamics during cirrhogenesis in rats. *IEEE Trans Biomed Eng* 2018;65(10):2311–2322.
- [15] Ho H, Yu HB, Bartlett A, Hunter P. An in silico pipeline for subject-specific hemodynamics analysis in liver surgery planning. *Comput Methods Biomech Biomed Engin* 2020;23(4):138–142.
- [16] Debbaut C, De Wilde D, Casteleyn C, Cornillie P, Van Loo D, Van Hoorebeke L, et al. Modeling the impact of partial hepatectomy on the hepatic hemodynamics using a rat model. *IEEE Trans Biomed Eng* 2012;59(12):3293–3303.
- [17] Audebert C, Bekheit M, Bucur P, Vibert E, Vignon-Clementel IE. Partial hepatectomy hemodynamics changes: experimental data explained by closed-loop lumped modeling. *J Biomech* 2017;50:202–208.
- [18] Golse N, Joly F, Nicolas Q, Vibert E, Line P-D, Vignon-Clementel I. Partial orthotopic liver transplantation in combination with two-stage hepatectomy: a proof-of-concept explained by mathematical modeling. *Clin Biomech (Bristol, Avon)* 2020;73:195–200.
- [19] Balzan S, Belghiti J, Farges O, Ogata S, Sauvanet A, Defosse D, et al. The “50–50 criteria” on postoperative day 5: an accurate predictor of liver failure and death after hepatectomy. *Ann Surg* 2005;242(6):824–828. discussion 828–9.
- [20] Mullen JT, Ribero D, Reddy SK, Donadon M, Zorzi D, Gautam S, et al. Hepatic insufficiency and mortality in 1,059 noncirrhotic patients undergoing major hepatectomy. *J Am Coll Surg* 2007;204(5):854–862.
- [21] Audebert C, Bucur P, Bekheit M, Vibert E, Vignon-Clementel IE, Gerbeau J-F. Kinetic scheme for arterial and venous blood flow, and application to partial hepatectomy modeling. *Comput Methods Appl Mech Engrg* 2017;314:102–125.
- [22] Hansen N, Müller SD, Koumoutsakos P. Reducing the time complexity of the derandomized evolution strategy with covariance matrix adaptation (CMA-ES). *Evol Comput* 2003;11(1):1–18.
- [23] Vilgrain V, Vullierme M, Djabbari M, Sibert A, Menu Y. Liver and vascular abnormalities. *J Radiol* 2002;83(2 Pt 2):255–268.
- [24] Wayson MB, Leggett RW, Jokisch DW, Lee C, Schwarz B, Godwin W, et al. Suggested reference values for regional blood volumes in children and adolescents. *Phys Med Biol* 2018;63(15):155022.
- [25] Bogner A, Reissfelder C, Striebel F, Mehrabi A, Ghamarnejad O, Rahbari M, et al. Intraoperative increase of portal venous pressure is an immediate predictor of posthepatectomy liver failure after major hepatectomy: a prospective study. *Ann Surg* 2019:1.
- [26] Truant S, Oberlin O, Sergent G, Lebuffe G, Gambiez L, Ernst O, et al. Remnant liver volume to body weight ratio $\geq 0.5\%$: a new cut-off to estimate postoperative risks after extended resection in noncirrhotic liver. *J Am Coll Surg* 2007;204(1):22–33.
- [27] Roayaie S, Jibara G, Tabrizian P, Park J-W, Yang J, Yan L, et al. The role of hepatic resection in the treatment of hepatocellular cancer. *Hepatology* 2015;62(2):440–451.
- [28] Sainz-Barriga M, Scudeller L, Costa MG, de Hemptinne B, Troisi RI. Lack of a correlation between portal vein flow and pressure: toward a shared interpretation of hemodynamic stress governing inflow modulation in liver transplantation. *Liver Transpl* 2011;17(7):836–848.
- [29] Bruix J, Castells A, Bosch J, Feu F, Fuster J, García-Pagán JC, et al. Surgical resection of hepatocellular carcinoma in cirrhotic patients: prognostic value of preoperative portal pressure. *Gastroenterology* 1996;111(4):1018–1022.
- [30] Iranmanesh P, Vazquez O, Terraz S, Majno P, Spahr L, Poncet A, et al. Accurate computed tomography-based portal pressure assessment in patients with hepatocellular carcinoma. *J Hepatol* 2014;60(5):969–974.
- [31] Annet L, Materne R, Danse E, Jamart J, Horsmans Y, Van Beers BE. Hepatic flow parameters measured with MR imaging and Doppler US: correlations with degree of cirrhosis and portal hypertension. *Radiology* 2003;229(2):409–414.
- [32] Buechter M, Manka P, Theyssohn JM, Reinboldt M, Canbay A, Kahraman A. Spleen stiffness is positively correlated with HVPG and decreases significantly after TIPS implantation. *Dig Liver Dis* 2018;50(1):54–60.
- [33] Rajakannu M, Cherqui D, Ciaccio O, Golse N, Pittau G, Allard MA, et al. Liver stiffness measurement by transient elastography predicts late post-hepatectomy outcomes in patients undergoing resection for hepatocellular carcinoma. *Surgery* 2017;162(4):766–774.
- [34] Fernández-Placencia R, Golse N, Cano L, Allard MA, Pittau G, Ciaccio O, et al. Spleen volumetry and liver transient elastography: predictors of persistent posthepatectomy decompensation in patients with hepatocellular carcinoma. *Surgery* 2020;168(1):17–24.

- [35] Bosch J, Abraldes JG, Berzigotti A, García-Pagán J-C. The clinical use of HVPG measurements in chronic liver disease. *Nat Rev Gastroenterol Hepatol* 2009;6(10):573–582.
- [36] Makuuchi M, Kosuge T, Takayama T, Yamazaki S, Kakazu T, Miyagawa S, et al. Surgery for small liver cancers. *Semin Surg Oncol* 1993;9(4): 298–304.